Statement ⇒

P:

Open sentence

Q: x is even , $Q(x)$

R (f,g): $f(x)$ is the derivative of the function $g(x)$

$\cos(x) = 1$

S: $\cos(x) = 1$, $S(x)$

Today:

Composing 2 statements:

R_1: the number 2 is even

R_2: the number 3 is odd

P: the number 2 is even and the number 3 is odd

combined statement: True

S_1: it is raining outside

S_2: we are doing MAT 111

P: $\underbrace{\text{it is raining outside}}_{S_1}$ and $\underbrace{\text{we are doing MAT 111}}_{S_2}$

$P: s_1 \wedge s_2$

$\wedge$: logical and,

The combined statement is false if Either one of the statement is false,

P	Q	$P \wedge Q$
T	T	T
T	F	F
F	T	F
F	F	F

Or:

s_1: the number 2 is even

s_2: the number 3 is odd

P: the number 2 is even or the number 3 is odd

so, combined statement is true if either one of them is true,

$P: s_1 \vee s_2$

P	Q	$P \vee Q$
T	T	T
T	F	T
F	T	T
F	F	F

P_1: 3 is even $\qquad$ P_2: 3 is odd

S: $P_1 \vee P_2$ $\rightarrow$ • P_1 or P_2 but not both

• Either P_1 or P_2

Negation:

P: 2 is even

$\sim P$: negation of the statement P

$\sim P$: 2 is not even

• excercise:

$\boxed{\text{Happy families are all alike}}$ but $\boxed{\text{each unhappy family is unhappy on their own way}}$

$\wedge$

$\boxed{\text{Human beings want to be good}}$ $\rightarrow S_1$

but not $\boxed{\text{too good.}}$ $\rightarrow S_2$: Human beings wants to be too good

$S_1 \wedge \sim(S_2)$

Conditional statement:

R: if P then Q

P: $\hat{a}$ is a multiple of 2 Q: $\hat{a}$ is divisible by 2

R: if P then Q

R: $P \Rightarrow Q$

Think of $P \Rightarrow Q$ as a promise

whenever P is true Q is true

- the promise will only be broken when P is true but Q is false

example:

P: you pass your exam	Q you pass the course	$P \Rightarrow Q$
T	T	T
T	F	F
F	T	T

Passing the exam is not the only

F way to pass the course F T

proof by contradiction

$P \Rightarrow Q$

F T F

$P \Rightarrow Q$

if P then Q

- If it rains (P), you will get wet (Q)
- $P \Rightarrow Q$ being true means that it is impossible for P be true and Q false

P	Q	$P \Rightarrow Q$
T	T	T
T	F	F
F	T	T
F	F	T

so, in order for P to be true it is necessary that Q is true.

Hence, "Q is necessary condition for P"

P only if Q

5: if you are a square (P), then you are a rectangle (Q)

if P then Q says the same thing as P only if Q

- you fail only if you stop writing

P Q

if you fail then you stop writing.

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one directional arrow:

$P \Rightarrow Q$

P	Q	$P \Rightarrow Q$
T	T	T
T	F	F
F	T	T
F	F	T

$Q \Rightarrow P$

Q	P	$Q \Rightarrow P$
T	T	T
F	T	T
T	F	F
F	F	T

(a is even) $\Rightarrow$ (a is divisible by 2)

(a is divisible by 2) $\Rightarrow$ (a is even)

$P \Rightarrow Q$ $\qquad$ $Q \Rightarrow P$

$$P \Leftrightarrow Q$$
$$\parallel$$
$$(P \Rightarrow Q) \wedge (Q \Rightarrow P)$$

$(P \Rightarrow Q)$: P only if Q

$(P \Leftrightarrow Q)$: P if and only if Q

P	Q	$P \Rightarrow Q$	$Q \Rightarrow P$	$(P \Rightarrow Q) \wedge (Q \Rightarrow P)$
T	T	T	T	T
T	F	F	T	F
F	T	T	F	F
F	F	T	T	T

$P \Rightarrow Q$ (Q is necessary for P to be true)

P: is not necessary condition for Q to hold

sufficient condition

$P \Leftrightarrow Q$: P is both necessary and sufficient condition for Q

- P is equivalent to Q

$x+3=5 \longrightarrow 0\cdot(\ )=0\cdot(\ )$

$\rightarrow 0=0$

$\rightarrow x+2=4$